KnowledgeDesk — Offline Help with Cloud AI & Local LLM — v1.2 Release Notes

Release Date: 16 October 2026 • **Version:** 1.2 • **Previous:** 1.0

🎉 What's New in v1.2

> 💡 **New Usage Tip — Graceful Shutdown:** The KnowledgeDesk server now runs minimized in your Windows taskbar. Closing this terminal window safely shuts down the server (and the browser will show a "Connection Refused" message). You can restart the app at any time from the desktop shortcut.

📚 Two Complete Companion Guides NEW

The installer now ships with two full HTML guides, both opened from the Finish page or accessible from the Help menu at any time:

- **USER-GUIDE.html** — interface tour, AI features, cloud and Local LLM configuration, reader tools, troubleshooting.
- **INSTALLATION-GUIDE.html** — system requirements, installer walkthrough, first-launch walkthrough, optional Cloud AI and Local LLM setup, verification and uninstall.

HIGHLIGHTS INSIDE THE GUIDES

- **Annotated "Summarize Topic" figure** — ① ② ③ ④ markers highlight the trigger button, the summary card, the two reset buttons, and the Copy button. Hover any marker for the full explanation.
- **Light Mode + Dark Mode** screenshots side by side, with a table explaining what changes between the two themes.
- **Copy summaries to another app** section — covers local summaries, AI Mode summaries, and Local LLM summaries, and explains exactly what "Copy" puts on the clipboard.
- Placeholder panels for screenshots still to be captured — no broken-image icons.
- **Print Topic:** Generates clean, unformatted PDFs of active topics. Perfect for sending specific instructions to clients or providing feedback to the documentation team without including UI navigation elements.
- 💬 **Enhanced Tooltips:** Added and refined hover tooltips across the entire interface. Hovering your mouse over buttons, fields, and icons now provides instant clarity on what they do, reducing the learning curve for new users.

✨ UI Polish & Custom Modals IMPROVED

We've replaced all native browser popups (the clunky grey `alert()` and `confirm()` boxes) with a beautiful, theme-aware custom modal engine. This affects:

- ⭐ **Favorites:** Adding a topic now shows a clean green "Success" modal. Clearing all favorites triggers a red "Confirm" modal with a Cancel button to prevent accidental data loss.
- 🔑 **API Keys & Local LLM:** All validation, save confirmations, and clear actions now use the same unified modal system.

- 🌓 **Light/Dark Mode:** All modals automatically adapt their colors, backgrounds, and text to match your current theme perfectly.

📚 Documentation & Troubleshooting Updates UPDATED

The Installation Guide and User Guide have been significantly expanded based on real-world testing to make setup and troubleshooting easier:

- **Graceful Shutdown:** Added instructions on how to properly close the minimized server window, and why the browser shows a "127.0.0.1 refused to connect" message when you do.
- **Local Storage Clarification:** Added clear explanations that API keys, bookmarks, and Local LLM settings are saved in the browser's Local Storage. Uninstalling the app does not wipe them, ensuring they survive future upgrades.
- **Local LLM Help:** Added step-by-step instructions in the User Guide on how to find the Ollama system tray icon and restart the service after setting the `OLLAMA_ORIGINS` environment variable.
- **Main Interface Tour:** Replaced the interactive image markers with a clean, accessible ordered list describing every button and tab in the UI.

🔊 Text-to-Speech — Expanded Voice Support NEW

The 🔊 **Read Topic** feature reads the active help topic aloud, with:

- Synchronized paragraph highlighting (the current paragraph is highlighted in yellow as it is spoken)
- Pause / Resume / Stop controls
- A voice selector dropdown in the navigation bar

VOICE AVAILABILITY BY BROWSER

BROWSER	VOICES	NOTES
Microsoft Edge	**~337 voices**	Male and female speakers across US, UK, Australia, Canada, India, South Africa, Ireland, New Zealand, and many non-English languages. Uses Microsoft Azure Neural voices built into Windows 10 / 11.
Google Chrome	**~25–40 voices**	Uses Google's online TTS engine. Much smaller catalog than Edge. Voice count varies by platform.
Brave	~25–40 voices	Same TTS source as Chrome.
Firefox	❌ **Not supported**	Does not expose the `speechSynthesis` API the way Edge and Chrome do.
Safari	Limited / inconsistent	WebKit implementation is not reliable.

> ✅ **Best experience:** Microsoft Edge on Windows 10 / 11 — **~337 voice variations** covering male and female speakers in dozens of regional accents. Chrome works but ships with a much smaller set of ~25–40 voices. If voice choice matters, use Edge.

For the best voice quality and the largest catalog, use **Microsoft Edge** on Windows 10 / 11. It uses the Microsoft Azure neural voices built into the OS.

> 💡 **Tip:** install additional language packs from *Windows Settings → Time & Language → Speech → Voices* to unlock even more voices. The app automatically picks up any voice your browser exposes.

🎤 Audio Input — Dictate Your Prompt NEW

The AI Assistant and the Advanced Search field now support **speech-to-text** via the browser's Speech Recognition API. Click the microphone button next to the prompt input, speak your question, and the transcribed text lands directly in the field. Combine with the AI Assistant for a hands-free "ask a question, get an answer" workflow.

WHERE THE MICROPHONE APPEARS

- 🤖 **AI Assistant panel** — next to the "*Type prompt here…*" textarea
- 💬 **Click Me popup** — next to the "*Ask AI or search web…*" input

The microphone is intentionally **not** attached to the Advanced Search field in the right pane — voice dictation is designed for AI prompts, not keyword search.

HOW IT WORKS

1. Click the 🎤 microphone button — it enters a recording state (visual pulse on the button).
2. Speak your question at normal pace. The browser transcribes as you go.
3. When you stop speaking, the recognized text is inserted into the prompt field.
4. Press `Enter` or click **Send Prompt** to submit.
5. Click the microphone again to cancel early if needed.

BROWSER SUPPORT

BROWSER	SUPPORT	NOTES
Microsoft Edge	✅ Full	Recommended on Windows
Google Chrome	✅ Full	Uses Google's cloud speech service
Brave	⚠️ Limited	May prompt for microphone permission differently
Firefox	❌ Not supported	Falls back gracefully — a message explains it
Safari	❌ Not supported	WebKit implementation is inconsistent

> 💡 **Tip:** The microphone button uses your browser's built-in speech recognition:
>
> - **Chrome** — audio is sent to Google for transcription.
> - **Edge** — audio is sent to Microsoft Azure for transcription.
>
> In both cases the audio leaves your computer for cloud transcription, and only the text result is returned. If you need fully offline voice input, it is not yet available — a fully local version is planned for v1.3 alongside Local LLM improvements.

🎛️ Renamed: Settings Panel

The settings panel from v1.0 has been split into two top-level panels, each with its own navigation button:

V1.0	V1.2	PURPOSE
`⚙️ AI & Search Settings` (single panel)	🔑 **Cloud AI & Web Search Keys**	Cloud provider API keys (Gemini, OpenAI, Claude, etc.) and web search keys (Bing)
	🖥️ **Local LLM (Offline)**	Offline AI engines (Ollama, LM Studio, LocalAI), hardware detection, and enable/disable toggle

> ℹ️ **Migration for existing users:** All cloud API keys saved in v1.0 are preserved automatically. No re-entry required. Simply access them under the new 🔑 **Cloud AI & Web Search Keys** navigation button.

🔍 Search Panel Changes

The Advanced Search panel has been updated to make searching faster and more reliable:

- Autocomplete & fuzzy typo correction (added in a prior release)
- Google Search API field removed — Google is discontinuing the Custom Search JSON API on 2027-01-01. The CSE widget continues to work without an API key.

🖥️ Local LLM (Offline AI) — Beta NEW

Run AI summaries and chat **entirely on your own machine**. No API key required. No data leaves your computer.

FEATURES

- Supports **Ollama**, **LM Studio**, and **LocalAI** engines
- Auto-discovers installed models from the engine
- Hardware detection (CPU / RAM / GPU) with tier-based recommendations
- Model Guide with size-class explanations and quality tiers
- Enable/disable toggle — **OFF by default** for safety
- Warm model retention (30 minutes) — subsequent calls stay fast
- Green 🖥️ badge shows the response came from your local model

HOW TO ENABLE

1. Install Ollama: ollama.com/download
2. Pull a model: `ollama pull llama3.2:3b`
3. Set environment variable: `OLLAMA_ORIGINS=http://127.0.0.1:39820`
4. Restart Ollama (quit from tray, relaunch)
5. In KnowledgeDesk: open 🖥️ **Local LLM (Offline)** tab
6. Click 🔄 **Discover Models** to find your models
7. Select a model (recommended: `llama3.2:3b`)
8. **Turn ON** the "Enable Local LLM" toggle
9. Use 📝 **Summarize Topic** or 🤖 **AI Assistant**

> ℹ️ **Why OLLAMA_ORIGINS?**
> Ollama blocks browser requests by default for security. The environment variable whitelists KnowledgeDesk's origin (`http://127.0.0.1:39820`) so the app can talk to it.

☁️ Extended Cloud AI Support

Grew from 4 providers (v1.0) to **11 providers** (v1.2):

TIER	PROVIDERS
Free	Google Gemini, Groq
Freemium	Mistral AI, Cohere
Paid	OpenAI, Anthropic Claude, DeepSeek, OpenRouter
Web Search	Bing API Key
Public ID	Google CSE ID (ships preconfigured)

GROQ IS NOW THE DEFAULT AI PROVIDER IMPROVED

KnowledgeDesk now defaults to **Groq** (Llama 3) out of the box, instead of Google Gemini.

- **Lightning Fast:** Groq provides incredibly fast inference speeds, meaning your summaries and chat replies appear almost instantly.
- **Reliability:** Google's free Gemini tier frequently experiences 503 "high demand" errors. Groq's free tier is much more stable for daily use.
- **Timeout Fallback:** If you have Local LLM enabled but your computer's hardware is too slow to generate a response within 150 seconds (causing a timeout), the app will gracefully fall back to Groq (if a key is saved) instead of failing completely.

GEMINI 3.8 FLASH SUPPORT UPDATED

Google's **Gemini 3.8 Flash** is now the primary model for AI Mode, with automatic fallback to 3.7 Flash and 3.6 Flash. The model cascade is:

```
gemini-3.8-flash → gemini-3.7-flash → gemini-3.6-flash
```

Users get the best available model on every request, and the app degrades gracefully if a newer model is temporarily unavailable in a region.

> ⚠️ **Deprecated models:** Gemini 3.5 Flash was deprecated by Google in October 2026 and is no longer called by this app. If you previously pinned 3.5 in a custom configuration, it will be ignored — the cascade starts at 3.8.

REMOVED DEPRECATED GEMINI PARAMETERS UPDATED

Gemini 3.6 and later **reject** `temperature`, `topP`, and `topK` in the request body. v1.2 removes these parameters from all Gemini API calls. AI Mode now returns clean responses on every Gemini Flash generation with no 400 errors in the console.

CENTRALIZED CLOUD MODEL REGISTRY UPDATED

Model lists, endpoints, and generation parameters for every cloud provider now live in one place inside `vault.js` (`VAULT.CLOUD_MODELS`). Future model updates — when Google ships 3.9 or 4.0 — become a one-line data change rather than a code change scattered across the app.

🔀 Three-Tier Response Routing

AI features automatically route through three tiers:

1. 🖥️ **Local LLM** — if enabled and a model is selected
2. ☁️ **Cloud AI** — if a cloud API key is saved
3. 📄 **Offline extract** — always available, no setup needed

Every response shows a colored badge indicating the source:

BADGE	MEANING
🖥️ Local LLM Summary	Generated by your local model
☁️ Assistant (AI Mode)	Generated by a cloud provider
📄 Offline Summary	Keyword extract (instant, no AI)

💡 AI Follow-up Suggestions Are Now Gated IMPROVED

The 💡 **Need Next Steps?** suggestion appears only when the app can actually answer a follow-up — i.e. when a Cloud AI key is saved or Local LLM is enabled. On a fresh install with no AI backend configured, the suggestion is replaced by a friendly **Unlock AI follow-ups** nudge pointing to the two setup paths.

This closes a class of confusing dead-ends where users typed "*Yes please?*" and got "*No active AI API key found.*"

EXPANDED "NO API KEY" HINT

When no AI key is stored, the AI Assistant now lists all supported providers grouped by tier — free, freemium, and paid — plus a direct pointer to the Local LLM (Offline AI) option. It also reminds you to reload the app after saving a key.

📋 Copy Buttons Preserve Formatting IMPROVED

The 📋 **Copy** button on AI responses now places *rich HTML* on the clipboard as well as plain text. Pasting into Word, Outlook, Teams, or OneNote preserves bold text, bullet lists, and headings. Pasting into a plain-text field still works because a plain-text fallback is included.

🎛️ Local LLM Enable/Disable Toggle

- Defaults to **OFF** on fresh install
- Persists across sessions via localStorage
- Disabling immediately falls back to cloud/offline
- Turning it back on resumes local LLM usage

> ⚠️ **Why default OFF?**
> New users who haven't installed a local LLM engine would see "Failed to fetch" errors on every summarize. Defaulting OFF lets users opt in *after* they've installed Ollama and verified the connection.

🔧 Go Launcher Improvements

- **Fixed port** at `127.0.0.1:39820` — no silent rolling
- Reuses existing instance if one is already running
- Clear error dialog if a foreign app owns the port
- Health endpoint at `/api/health` for instance detection

> ✅ **Why no port rolling?**
> Rolling to a new port changes the browser origin (`127.0.0.1:39820` → `127.0.0.1:39821`), which makes all locally saved API keys disappear. A fixed port keeps everything stable across restarts.

📊 AI Replies Now Format Tables and Checkboxes NEW

AI-generated markdown tables previously appeared as raw pipe characters (e.g. `| Field | Description |`). This made dense field-by-field answers from Groq, Cohere, and other providers hard to read. v1.2 adds a markdown table renderer that converts them into proper HTML tables.

BEFORE AND AFTER

Before (v1.0 — raw pipes as plain text):

```
| Field        | What to enter                              |
|--------------|--------------------------------------------|
| Contact Name | Full name of the person or entity          |
| Account      | Choose the debtor or creditor account      |
| Default      | Tick to auto-select on sales/purchase docs |
```

After (v1.2 — real HTML table):

FIELD	WHAT TO ENTER
Contact Name	Full name of the person or entity
Account	Choose the debtor or creditor account
Default	Tick to auto-select on sales/purchase docs

WHAT THE TABLE RENDERER SUPPORTS

- **Header row** with shaded background
- **Separator row** detection (`|---|---|`)
- **Multiple body rows** — unlimited
- **Bold, italic, and code** markers inside cells
- **Short rows** padded with empty cells for alignment
- **Leading whitespace** tolerated on any line
- **Cells with pipe characters** in `` `code` `` blocks work correctly

THEME-AWARE STYLING

Table colors use the same CSS variables as the rest of the app, so tables automatically match the active theme:

ELEMENT	LIGHT MODE	DARK MODE
Table background	White	Charcoal
Header row	Light slate	Dark slate
Borders	Light grey	Medium grey
Body text	Near-black	Off-white

No hardcoded colors — the same table renders correctly whether you are in **Light Mode** or **Dark Mode**.

TASK-LIST CHECKBOXES

Related to tables, the AI sometimes returns task-list syntax:

```
- [ ] Open Contacts via Home ➔ Contacts.
- [ ] Add any new contacts needed.
- [x] Save the Chart of Accounts.
```

These now render as checkbox glyphs instead of literal brackets:

☐ Open Contacts via Home ➔ Contacts.

☐ Add any new contacts needed.

☑ Save the Chart of Accounts.

AI-generated checklists are now directly usable as to-do lists in the reply pane.

COPY PRESERVES TABLE STRUCTURE

Clicking 📋 **Copy** on a reply that contains tables places both HTML and plain text on the clipboard. Pasting into Word, Outlook, or Teams preserves the table structure as a real table. Pasting into a plain-text field produces a readable plain-text fallback.

> ✅ **Which AI providers benefit most?** Providers that favor structured output — **Groq**, **Cohere**, and **Mistral** — produce the most tables. Cloud summaries from these providers are now substantially more readable than in v1.0, where the same content rendered as raw pipes.

🎯 Summarize Topic — Unified Three-Mode View NEW

The 📝 **Summarize Topic** button now produces a single summary regardless of the underlying engine, but the header clearly identifies which mode handled the request:

HEADER	SOURCE
`🖥️ Local LLM Summary: [Topic]`	Generated by your local model — model name and elapsed time shown below the header
`📝 AI Summary (PROVIDER): [Topic]`	Generated by the named cloud provider (GEMINI, GROQ, OPENAI, DEEPSEEK, MISTRAL, COHERE, OPENROUTER)
`📄 Offline Summary: [Topic]`	Extractive summary from the local help index — no AI required

📃 Accounting-Focused System Prompt IMPROVED

All cloud AI and local LLM prompts now carry a system instruction that establishes the assistant as a TurboCASH/osFinancials technical documentation helper. The prompt:

- Requests concise, practical answers with exact menu paths
- Prefers bullet points and numbered steps over long prose
- Expands into a full walk-through only when the user asks for one
- Keeps replies to 2–5 sentences for simple questions and up to 5–7 numbered steps for "how do I" questions
- Reminds the model to bold software commands and menu paths

🔍 Key Management — Validation and Filters NEW

The four action buttons in the 🔑 **Cloud AI & Web Search Keys** panel now validate and report comprehensively:

BUTTON	CHANGE
💾 **Save Keys**	Validates every key on save. Reports a green **Success** dialog when all keys pass format checks, or an amber **Warning** dialog that lists valid and invalid keys separately.
⚡ **Test AI Keys**	Reports the status of all 8 AI providers — including ones with no saved key. Shows *Valid* ✅, *Invalid format* ⚠️, or *Missing* ❌ for each.
🌐 **Test Web Keys**	Reports the status of Bing and Google CSE credentials.
🗑️ **Clear All Keys**	Now shows a confirmation dialog that lists every provider whose keys will be cleared, plus the current default model — before the clear is executed.

CROSS-PROVIDER PREFIX VALIDATION

The validator now rejects keys clearly belonging to a different provider. A Groq key (`gsk_...`) saved in the Gemini field reports **Invalid format** ⚠️ — catching paste mistakes before they cause failed requests later.

🗑️ Per-Key Delete — Safe Confirmation FIXED

The individual 🗑️ buttons in the API key fields previously used the browser's native `confirm()` dialog and could remove the key before the user confirmed. They now use the app's modal dialog system with **Cancel** and **Delete** buttons, and the key is only removed when **Delete** is clicked.

🎨 Consolidated Modal Dialogs IMPROVED

Every notification in the Cloud AI & Web Search Keys panel now uses the same styled modal dialog:

- Centered on screen with a dimmed backdrop
- Color-coded by type — green (success), amber (warning), blue (info), red (error)
- Dismissible with the **OK** button, `Esc`, or by clicking outside
- Scrollable if content overflows

No native browser popups remain anywhere in the settings panel.

📊 Markdown Table and Checkbox Rendering NEW

AI replies that contain pipe-delimited markdown tables now render as proper HTML tables with theme-aware colors — dark header backgrounds and borders in dark mode, light variants in light mode. Task-list syntax `[ ]` and `[x]` renders as checkbox glyphs (☐ and ☑).

🔒 Key Format Refinements FIXED

The Gemini key validator now accepts both `AIza...` (legacy) and `AQ....` (newer) formats. A follow-up refinement prevents the loosened validator from accepting keys with prefixes that belong to other providers.

🐞 Bug Fixes

Removed Deprecated `temperature` from Gemini Requests FIXED

Gemini 3.6+ rejects `temperature`, `topP`, and `topK`. Sending them caused a silent 400 error that appeared as "AI service unavailable" in the UI. v1.2 strips these parameters from every Gemini request.

Duplicate `bing` Provider Entry FIXED

The provider registry contained a stub `bing` definition that was silently overwritten by a complete one. Removed the dead entry — no functional change, but the file is now cleaner and won't trip up future edits.

Corrected Provider Resolution in `activeAIProviders()` FIXED

The function referenced a non-existent `'googleSearch'` provider, which prevented the filter list from being accurate. v1.2 filters only on real providers.

Follow-up Prompts No Longer Dead-End FIXED

Previously, the app suggested a follow-up question (*"Would you like step-by-step guidance on…?"*) even when no AI backend was configured. Clicking it produced the *"No active AI API key found"* message. Follow-up suggestions are now suppressed in that state, and a setup nudge appears instead.

Markdown Rendering Preserves Menu Arrows FIXED

Menu paths copied from HelpNDoc, such as `Reports → Currencies`, now render with heavy arrows (`➔`) that stay visible at small font sizes, instead of the thin Unicode arrow that got lost in dark mode.

📋 Copy Buttons Preserve Formatting & Context IMPROVED

The 📋 **Copy** button on AI responses now places *rich HTML* on the clipboard as well as plain text. Pasting into Word, Outlook, Teams, or OneNote preserves bold text, bullet lists, and headings. Pasting into a plain-text field still works because a plain-text fallback is included.

SOURCE HEADERS & MODEL INFO INCLUDED NEW

When you copy a summary or chat reply, the clipboard now automatically captures the **Source Header**. This means the text you paste will include the engine used (Cloud AI, Local LLM, or Offline Extract), the topic name, and if applicable, the specific local model name and generation time.

Example of what gets pasted:

> 🖥️ Local LLM Summary: Contacts
> qwen2.5:3b • 23.8s
>
> The Contacts feature in osFinancials5/TurboCASH5 allows users to maintain...

This ensures that when you send a summary to a client, colleague, or support ticket, the recipient knows exactly where the information came from.

Copy Button No Longer Overlaps Reply Text FIXED

The 📋 **Copy** button that appears on every AI reply is positioned absolutely at the bottom-right of the response card. In v1.0, on replies with several lines of text, the button would sit on top of the last paragraph, hiding the final words.

The reply container now reserves a fixed amount of bottom padding (`padding-bottom: 40px` on `.ai-response-body`), so the last line of any reply always sits clear above the button. The fix applies uniformly to:

- Cloud AI summaries (`AI Summary (PROVIDER):`)
- Local LLM summaries (`Local LLM Summary:`)
- Offline extracts (`Offline Summary:`)
- Chat replies (`Assistant (AI Mode · PROVIDER):`)
- Basic-mode chat results (`Assistant (Basic Mode Results):`)

The fix is purely CSS — one rule at the end of `viewer.css` — so it costs no additional code in the response logic and cannot regress if the reply HTML is updated later.

🗑️ Files Removed in v1.2

REMOVED	REASON
`port.json`	Legacy bookkeeping file. The Go launcher no longer needs it — health-check on `/api/health` replaced it.
`webview2-diagnostic.html`	Test page from an earlier WebView2-based architecture. No active code referenced it.
`webview2-diagnostic.js`	Diagnostic script that went with the above. Orphaned.
Google Search API Key field	Google is discontinuing the Custom Search JSON API on 2027-01-01 and no longer issuing new keys. The CSE widget continues to work without any API key.

📁 Files Changed

FILE	V1.0	V1.2	CHANGE
`index.html`	623 lines	1,126 lines	+503
`viewer.js`	5,372 lines	~7,062 lines	+1,690
`viewer.css`	1,687 lines	1,788 lines	+101
`vault.js`	—	1,143 lines	NEW
`main.go`	—	211 lines	NEW
`USER-GUIDE.html`	—	new	NEW
`INSTALLATION-GUIDE.html`	—	new	NEW
`ai-config.json`	7 keys / 9 lines	10 keys / 12 lines	+3 keys
`port.json`	4 lines	deleted	REMOVED
`go.mod`	1 file	deleted	REMOVED
`webview2-diagnostic.html` `webview2-diagnostic.js`	2 files	deleted	REMOVED

Line counts include blank lines and comments.

Binary Size Change

BINARY	SIZE	BUILT	NOTES
Old `server.exe`	9,948,672 bytes	2026-09-04	Stale — wrote `port.json`
New `server.exe`	10,682,368 bytes	2026-10-09	Clean — no `port.json`, 32-bit build

⚠️ Known Limitations

- **Four cloud providers are visible but not yet routed:** Anthropic Claude, Mistral AI, Cohere, and OpenRouter keys can be saved, but AI Mode currently routes only to Gemini, Groq, OpenAI, and DeepSeek. Full routing arrives in v1.3. Workaround: use Gemini or Groq (both free), or enable Local LLM.
- **First cold call after Ollama starts** takes 60–140 seconds as the model loads into RAM. Subsequent calls are 5–20 seconds for up to 30 minutes.
- **Browsers round RAM** to the nearest power of 2 for privacy reasons. Use 🖍️ **Edit Profile** in the Local LLM tab to enter exact RAM.
- **Small models (3B)** may approximate exact menu paths. Use a larger model (7B+) for higher precision.
- **No conversation memory** — each chat message is independent. A follow-up like "Yes please" won't have context from the previous message.
- **Speech recognition uses Google's cloud service** in Chrome and Edge. Audio is sent to Google for transcription, not processed locally.
- **Web grounding is optional** — requires a Bing API key. The app works fine without it.

🔬 Debug & Diagnostics

This section is for testers and users who want to verify the installation or troubleshoot a specific area. Every check below can be run from the browser's Developer Tools console (`F12`).

1. Open Developer Tools

Press `F12` (or `Ctrl` + `Shift` + `I`) in Chrome, Edge, or Brave. The DevTools panel opens docked to the side or bottom. Switch to the **Console** tab.

2. Verify the Vault Loaded

On every page load, the vault logs a single startup line. Look for a line beginning with `[Vault]` :

> `[Vault] Origin: http://127.0.0.1:39820 | Storage: available | User keys: N | CSE ID: 4753ce97761ac4f9d | Local: llama3.2:3b (disabled)`

READING THE LOG

FIELD	MEANING	GOOD VALUE
Origin	Where the app is served from	`http://127.0.0.1:39820`
Storage	Whether localStorage is usable	`available`
User keys	How many API keys you've saved	`0` or higher
CSE ID	Google Custom Search Engine identifier	The built-in default ships preconfigured
Local	Which Local LLM model is remembered	Any name, followed by `(disabled)` if the toggle is off

3. Verify Cloud Model Registry

Confirm the Gemini model cascade and generation config. Run this in the console:

```
VAULT.CLOUD_MODELS.gemini.models
// Expected: ['gemini-3.8-flash', 'gemini-3.7-flash', 'gemini-3.6-flash']

VAULT.CLOUD_MODELS.gemini.defaultConfig
// Expected: { maxOutputTokens: 8192 }    (no temperature field)
```

4. Verify Provider Registry

Confirm the provider registry loads without duplicates:

```
Object.keys(VAULT.PROVIDERS).filter(k => k === 'bing').length
// Expected: 1

Object.keys(VAULT.PROVIDERS).length
// Expected: 10  (8 AI providers + 1 web search + 1 public ID)
```

5. Verify an API Key Is Stored Correctly

After pasting a Gemini key in the settings panel and clicking 💾 **Save Keys**, run:

```
VAULT.activeProviders()
// Expected: ['gemini'] if you saved one key

VAULT.validate('gemini', VAULT.get('gemini'))
// Expected: true if the key starts with AIza and is at least 35 chars

VAULT.mask(VAULT.get('gemini'))
// Expected: something like AIza••••••••••••3f1x
```

6. Test the AI Request Path

With a key saved, click 📝 **Summarize Topic** on any help topic, then check the Network tab:

1. Open DevTools → **Network** tab
2. Filter by `generateContent`
3. Click the request → **Payload** tab
4. Confirm the URL contains `gemini-3.8-flash`
5. Confirm the payload has `maxOutputTokens: 8192` and **no** `temperature` field
6. Confirm the response arrives with a valid summary

7. Audio Input — Dictate Prompt to Microphone

Verify the 🎤 microphone button works in your browser.

PRECONDITIONS

- Use **Chrome** or **Edge** — Firefox and Safari do not support the Web Speech API in the way this feature uses it.
- The app must be loaded over `http://127.0.0.1:39820` or `https` — browsers block microphone access on insecure origins other than localhost.
- Your operating system must grant the browser access to the microphone.

WINDOWS MICROPHONE PERMISSION

1. Open **Settings** → **Privacy & Security** → **Microphone**
2. Confirm **Microphone access** is ON
3. Confirm **Let apps access your microphone** is ON
4. In the app list, ensure **Google Chrome** or **Microsoft Edge** is enabled

TESTING THE MICROPHONE

1. Click 😈 **AI Assistant** in the navigation bar
2. Click the 🎤 **microphone button** next to the prompt input — it should enter a visible "recording" state
3. Say "*What is a journal entry?*" at a normal pace
4. The transcription should appear in the prompt field within a second of you stopping

5. Press `Enter` to submit

IF THE MICROPHONE DOESN'T ACTIVATE

SYMPTOM	CAUSE	FIX
Alert: "Speech Recognition is not supported in this browser"	Firefox or Safari	Switch to Chrome or Edge
Button stays inactive	Browser hasn't been granted mic permission	Click the 🔒 padlock in the address bar → Site settings → allow Microphone
Console: `not-allowed`	Microphone blocked at OS level	See "Windows microphone permission" above
Console: `no-speech`	Silence detected during the recording window	Speak sooner after clicking, and check the correct input device is selected
Console: `network`	Cloud speech service unreachable	Check your internet connection — speech recognition is cloud-based

> 💡 **Console evidence:** when the microphone activates successfully, you will see `onstart` in the console. When transcription completes, the recognized text is inserted into the prompt field. If nothing appears, check for `onerror` events with the codes listed above.

8. Local LLM Connection Check

Confirm Ollama (or another local engine) is reachable:

```
// From the browser console (Ollama default):
fetch('http://localhost:11434/api/tags')
  .then(r => r.json())
  .then(d => console.log(d.models.map(m => m.name)))
// Expected: an array of installed model names

// From PowerShell (Windows):
Invoke-WebRequest http://localhost:11434/api/tags
```

9. Reset Stored Configuration

If keys, favorites, or preferences need to be wiped clean:

```
// Clear all API keys (keeps favorites and Local LLM preferences):
VAULT.clearAll()

// Full reset — clears everything this app stores:
localStorage.clear()
location.reload()
```

> ⚠️ **Warning:** `localStorage.clear()` wipes *all* data stored by this app for the current origin, including favorites, saved API keys, and Local LLM preferences. Export any key backups first if you need them.

10. Understanding the Log Messages

LOG LINE	MEANING
`[Vault] Origin: … | Storage: available | User keys: 0`	Vault initialized cleanly, no keys saved yet
`[Vault] Origin: … | Storage: UNAVAILABLE`	localStorage blocked (incognito mode or strict privacy) — keys won't persist
`[Vault] Active providers: gemini, groq`	Two keys are stored and active
`[Compat] displayOutput() — using shim`	Legacy code path — harmless, informational only
`[AI] Routing to local LLM: llama3.2:3b`	Request is being served by your local model, not the cloud
`[Summarize] Local LLM failed, falling back: …`	Local engine didn't respond — cloud AI or offline extract will handle it

✅ Testing Checklist for Testers

- ☐ Fresh install → Local LLM toggle is **unchecked**
- ☐ Enable toggle → select a model → **Summarize Topic** works with 🖥️ badge
- ☐ Disable toggle → **Summarize Topic** falls back to 📄 offline badge
- ☐ Add cloud API key → Summarize uses ☁️ cloud AI (check Network tab shows `gemini-3.8-flash`)
- ☐ Gemini request payload contains `maxOutputTokens: 8192` and **no** `temperature`

- [ ] Console shows `[Vault] Origin: http://127.0.0.1:39820 | Storage: available | User keys: N`
- [ ] CSE widget in 🌐 Google Web tab returns results
- [ ] Dark mode toggle works (header button)
- [ ] Read Topic (text-to-speech) works with pause/resume/stop
- [ ] Print Topic produces a clean printable page
- [ ] 🎤 Microphone button appears in AI Assistant; clicking it enters recording state in Chrome/Edge
- [ ] Console has no errors at startup

💻 System Requirements

- Windows 10 or 11 (32-bit or 64-bit)
- Modern browser — **Edge or Chrome recommended** for full Audio Input support; Firefox works but without microphone input
- **Optional for Local LLM:** Ollama installed and running

🚀 What's Next — v1.3

PLANNED	DESCRIPTION
AI Agents	Multi-step task automation built on the existing AI infrastructure — chain prompts, use the help index as a knowledge base, execute a workflow end-to-end (e.g. "compare two topics and produce a summary").
Offline voice input	Add a fully local speech-to-text option alongside the cloud one, using Whisper or a similar on-device model. This makes dictation available in Firefox and Safari, and keeps audio private — nothing leaves the machine.
Help Index — SEO & keyword review	A dedicated audit of the HelpNDoc-generated index (~762 keywords exposed to the app). Reconcile legacy terminology still used by long-standing TurboCASH / osFinancials users with modernised accounting-standard language (IFRS, GAAP, local tax regimes). Add synonym and alias entries so either term reaches the same topic. Improves both search relevance and AI summarization grounding in one pass, since both consume the same index.
Expanded Local LLM support	Extend the local engine list, tune hardware recommendations based on real-world telemetry, and add GPU-accelerated detection improvements for Ollama and LM Studio.
Provider health dashboard	A small status panel showing which providers are currently reachable, average response times, and which models in each fallback chain have recently succeeded or failed — visible at a glance rather than only in the console.

📮 Feedback

Report bugs, unclear UX, or performance issues to: osfinancials1@gmail.com

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